

IO SKY

IVR & Telephony

Functional Specification

Developer Implementation Specification · Version 1.0

Status: FINAL FOR IMPLEMENTATION

Scope: inbound telephony, AI Receptionist entry/fallback, traditional IVR, routing, queues, transfers, callbacks, voicemail, recording/transcription interfaces, SIP/business identity, administration, permissions, audit, analytics, concurrency and acceptance testing.

AUTHORITATIVE RELATIONSHIP

This document is the authoritative specification for the IVR and shared telephony layer. It must be implemented together with the IO SKY AI Receptionist Functional Specification v1.0 and the IO SKY AI Receptionist Conversation & Response Playbook v1.0. Where this document references AI conversation behaviour, the two AI Receptionist documents remain authoritative. This document does not replace or weaken their requirements.

Core principle

The AI Receptionist is the normal inbound caller experience. A caller must not hear the traditional DTMF menu merely because IVR capability exists. Traditional IVR is a controlled telephony fallback and routing capability.

1. Purpose, authority and interpretation

This specification removes implementation ambiguity from the IO SKY IVR and telephony layer. Requirements described as must, shall, required, never, only or non-negotiable are mandatory acceptance requirements. Examples illustrate intended behaviour unless explicitly identified as canonical caller-facing wording.

A developer must not substitute a materially different flow because a technical provider has a different default. If a provider limitation prevents a requirement, it must be raised before implementation acceptance with the limitation, impact and proposed compliant alternative documented.

- AI Receptionist Functional Specification: authority for AI capabilities, permissions, call intelligence and action boundaries.
- Conversation & Response Playbook: authority for conversational behaviour, tone, clarification and caller-facing intelligence.
- This IVR & Telephony Specification: authority for telephony entry, DTMF fallback, call routing, queue/hold, transfer, callback, voicemail, availability, business identity, telephony administration, concurrency and telephony failure handling.

NO IMPLIED OMISSIONS

A feature is not considered optional merely because the exact vendor, API, codec, SIP implementation or infrastructure mechanism is left to the development team. The required outcome and acceptance behaviour in this specification remain mandatory.

2. Launch operating model

The launch model is AI-first. The AI Receptionist is intended to be available 24/7, subject to provider and infrastructure availability. Human availability is separate from AI availability.

Normal call path:

Inbound caller → IO SKY business number → telephony entry layer → AI Receptionist session.

When the AI Receptionist is operationally available, the caller must enter the AI Receptionist directly and must NOT first hear a language-selection or DTMF department menu.

Fallback call path:

Inbound caller → telephony entry layer detects that the conversational AI path is unavailable or administratively disabled → traditional IVR fallback → language selection where required → DTMF menu → routing/fallback.

The existence of the IVR must never make the normal AI-first experience slower or force every caller through a keypad tree.

3. AI availability and IVR activation states

- AI AVAILABLE: route directly to an independent AI Receptionist session; no traditional DTMF menu.
- AI UNAVAILABLE: automatically enter traditional IVR if the telephony layer remains available.
- AI ADMINISTRATIVELY DISABLED: enter traditional IVR for the affected number/configuration.
- AI FAILS DURING AN ACTIVE CALL: where the remaining telephony layer is functional, recover the caller into a controlled fallback rather than silently disconnecting.
- AI RECOVERS WHILE A CALLER IS ALREADY IN IVR: do not abruptly move the active caller into AI. Keep the current IVR session consistent; new calls may use AI again.
- TOTAL TELEPHONY PROVIDER FAILURE: behaviour is constrained by the provider, but failures must be logged/alerted where technically possible and no application state may falsely claim a successful call action.

AI availability state and human availability state must never be conflated. The AI can be available while no human operator is available.

4. Language handling

The AI Receptionist automatically handles Dutch and English according to the AI Receptionist specifications. The traditional IVR uses a language selection only when the caller has entered the IVR fallback and the system cannot otherwise reliably select the appropriate configured IVR language.

Launch fallback language prompt may be implemented as: “Voor Nederlands, druk 1. For English, press 2.” The exact audio must be manageable by Super Admin.

Language selection applies to the current IVR session. The underlying routing/security rules must not be duplicated as separate uncontrolled Dutch and English implementations.

5. Launch DTMF menu - frozen caller-facing wording

DUTCH - CANONICAL LAUNCH WORDING

“Voor vragen over onze oplossingen of wat IO SKY voor u kan betekenen, druk op 1.
Voor een lopend project of als u al klant bent, druk op 2.
Voor support, druk op 3.”

ENGLISH - CANONICAL LAUNCH WORDING

“For questions about our solutions or how IO SKY can help your business, press 1.
For an ongoing project or if you're already an IO SKY client, press 2.
For support, press 3.”

The three caller-facing options are deliberately broader than internal classifications. Billing/administration and privacy/security are not separate launch menu options. They may be handled through the appropriate existing-client/support route and then internally classified or escalated.

A repeat-menu function must exist, but its key and spoken placement are configurable. It must not create an unnecessarily long primary menu.

6. Internal route meaning

Route	Intended scope	Examples
1 - Solutions / what IO SKY can do	New prospects and general commercial/business enquiries	Solutions, services, AI Scans, Discovery Call, potential collaboration, a caller who does not yet know the correct next step.
2 - Projects & Existing Clients	Existing relationships and ongoing delivery	Current project, delivery question, existing client matter, project-related administration or other relationship context.
3 - Support	Support and issues requiring operational assistance or escalation	Technical/support problem; privacy/security matter that requires human handling; other support issue.

7. IVR menu configurability

The launch wording above is frozen for initial implementation, but the IVR must not be hardcoded. Super Admin must be able to manage day-to-day IVR configuration without source-code changes or developer intervention.

- menu options, keys and order;
- Dutch and English audio/text per option;

- repeat-menu behaviour;
- route destination(s);
- active/inactive state;
- business-hours and exception behaviour;
- fallback route;
- queue/hold configuration and hold audio;
- routing priority and destination availability rules;
- callback/voicemail behaviour;
- approved recording/transcription settings and notices where applicable.

8. Invalid input, no input and loop prevention

The IVR must handle invalid/no DTMF input without trapping the caller. The caller may receive a natural retry. A second unsuccessful attempt must move to a configured safe fallback such as callback/voicemail/general support handling rather than looping indefinitely. Exact retry count and timeout values must be configurable within safe provider limits.

The system must distinguish no input, invalid key, provider DTMF failure and caller hang-up in operational logs where the provider exposes these states.

9. Human availability model

Configured business hours at launch are Monday-Friday 09:00-17:00 Europe/Amsterdam for potential human transfer. These hours do not guarantee that a human is currently available.

Effective availability must consider at least the configured operator state and, where integrated/authorised, relevant scheduling occupancy. A Super Admin must be able to mark themselves Available, Busy or Unavailable. A scheduled Discovery Call or other configured occupied state may make a destination unavailable without exposing the private reason to the caller.

CONTEXT IS FOR INTELLIGENCE, NOT NARRATION

The system may use availability context to avoid a pointless transfer attempt, but must not unnecessarily tell the caller internal details such as “the operator is currently in a Discovery Call.” A suitable response is that nobody is currently available to take over and a callback can be requested.

Super Admin must be able to configure holiday closures, exceptional hours, temporary unavailability and future after-hours/on-call destinations without code changes.

10. Routing engine

Routing must be destination- and availability-aware. The architecture must support more than the current single operator so IO SKY can add staff without rebuilding the telephony foundation.

- Primary destination → secondary destination → queue/fallback.
- Destinations may represent an authorised person, team, telephone number, SIP endpoint or other provider-supported business destination.
- Routing priority must be configurable.
- Availability may prevent an unavailable destination from being attempted.
- Future round-robin or other multi-operator strategies may be enabled without replacing the underlying call model.
- Internal category must remain attached to the call even when multiple categories currently terminate at the same human operator.
- Routing changes are versioned/audited and controlled by Super Admin.

11. Queue capability

A real queue capability is required even though launch may initially have one human operator. The implementation must not assume that IO SKY will remain a one-person receiving operation.

- Queue entry must be explicit in call state.
- A caller must not remain indefinitely in a queue.
- Queue limits, maximum wait/fallback conditions and destination behaviour must be configurable.
- Future multiple operators must be able to receive queued calls according to configured routing/availability.
- Abandoned queue calls must be distinguishable from failed transfers and normal caller hang-ups.
- When queue capacity or configured wait conditions are exceeded, the caller must move to a controlled fallback where technically available.

12. Hold music and waiting experience

Hold audio must be premium, calm and businesslike. No advertising, AI Scan promotion, sales messages or promotional interruptions are to be inserted into hold by default.

Super Admin must be able to upload/replace approved hold audio without developer intervention. Optional waiting/status messages may exist but must be sparse and configurable; repetitive messages every few seconds are expressly not acceptable.

The caller must never hear internal operator audio before the transfer has been correctly answered/bridged.

13. AI-to-human transfer preparation

A human transfer is not a blind forward. Before the receiving operator is expected to answer, the system must make a compact handoff summary available through the authorised operator interface and configured notification channels.

- caller name, if known and appropriate;
- organisation/company, if relevant;
- reason for the call;
- key context already established;
- relevant AI actions already completed;
- requested outcome / why human involvement is needed;
- urgency classification where relevant and permitted;
- current call/transfer state.

The summary must be factual and concise. It must not invent facts, expose hidden reasoning, or dump an entire transcript. The receiving human should be able to understand who is calling and why before or as the call is presented.

14. Immediate handoff notification and multi-device access

The transfer context must not depend on the operator already having the IO SKY dashboard open on a laptop. An immediate in-platform notification is required, plus support for configurable authorised external/mobile notification channels or secure push mechanisms.

The implementation must be provider-neutral: IO SKY may later choose a supported business channel such as mobile push or an approved collaboration/messaging connector. Sensitive information must be minimised in external notifications; where appropriate, the external notification should contain only necessary context plus a secure deep link to the authorised platform record.

A delivery failure on an optional external channel must not corrupt the call state or prevent the in-platform handoff context from existing.

15. Transfer announcement and ring window

The AI Receptionist must announce that it will attempt a transfer before initiating it, in accordance with the Conversation Playbook. The system must not imply that the human has already answered.

The transfer/ring attempt window must be realistically long enough for an authorised operator to receive the alert, identify the caller, read the compact summary and answer. It must be configurable within provider-safe limits. A very short hardcoded timeout that immediately converts a transfer into a callback is not acceptable.

If the human answers, the human may briefly place/keep the caller in a controlled hold state while reviewing the provided context, then greet the caller naturally. The caller should not be forced to repeat the entire story.

16. Transfer success and failure recovery

Transfer state must be explicit and server/provider-confirmed. Required outcomes include: attempted, ringing/presented where available, answered/bridged, no answer, busy, rejected, technical failure, cancelled and caller disconnected where supported.

If a transfer fails, the caller must not be silently disconnected. Where the AI conversational layer remains available, the AI must regain control and explain naturally that the transfer did not succeed, then offer a context-appropriate fallback. If AI is unavailable, the telephony/IVR layer must offer the configured fallback.

A failed transfer must never be recorded as successful.

17. Callback-first fallback

When a human is unavailable or a transfer fails, callback is the preferred structured fallback. Voicemail is secondary.

The system must not promise an exact callback time unless an explicit authorised workflow/SLA supports that promise. Default behaviour is to confirm that a callback request has been recorded.

A callback record must link to the relevant call/person/lead/client context where safely matched and must generate an actionable notification. Suggested lifecycle: New → Seen → In Progress → Resolved.

18. Caller-ID-aware callback flow

If a usable inbound caller ID is available, the caller must not be unnecessarily required to re-enter the same number. The IVR should confirm whether the caller wants to be called back on the number from which they are calling.

Example behaviour: “Wilt u teruggebeld worden op het nummer waarmee u nu belt? Druk 1 voor ja. Druk 2 om een ander nummer op te geven.” English equivalent must be configurable.

If caller ID is unavailable/withheld or the caller selects another number, the system may collect a callback number through the configured flow and confirm it before creating the request.

SECURITY BOUNDARY

Caller ID may assist contact/routing context. Caller ID alone is never proof of identity and must not unlock sensitive client actions.

19. Voicemail

Voicemail must be available as a configured fallback after/alongside callback options. A voicemail must create a durable call-related record and notification if successfully stored.

Voicemail transcription and summarisation may be separately configurable where legally and technically approved. A voicemail must never be marked saved if provider/backend confirmation failed.

20. Business-hours and after-hours behaviour

The AI Receptionist remains the intended 24/7 front door. Human transfer availability is governed separately. Outside configured human hours, the system must not knowingly attempt the normal human destination unless an authorised after-hours/on-call destination is configured.

The AI may continue all approved non-human actions outside hours, including answering approved questions, appropriate Discovery Call booking and callback capture. Traditional IVR fallback outside hours must provide the configured callback/voicemail/contact route.

Europe/Amsterdam is the launch authoritative timezone for human availability and business-hour evaluation.

21. Call recording state

Audio recording is OFF by default. Recording and transcription are separate controls.

When an authorised Super Admin activates recording for a legally validated/configured purpose, the active telephony stack must treat recording=ON as an operational state, not merely a database setting. The system must automatically apply the configured notice behaviour before relevant recording begins, logging, storage controls, access controls, retention handling and downstream processing required by the approved configuration.

The AI Receptionist must receive the authorised state when that state changes caller-facing behaviour. No generic or invented recording purpose may be announced.

22. Transcription and real-time speech processing

Real-time speech-to-text used by the AI to understand a caller does not automatically mean a permanent transcript is stored. Persistent transcription must be independently governed/configured.

Recording OFF does not necessarily mean real-time AI speech processing is OFF. Persistent transcript storage must not be inferred from the existence of STT.

Where persistent transcription or voicemail transcription is enabled, its access, retention, notices and storage must follow the approved privacy/security configuration.

23. Structured call summary

The durable default conversational artifact is a structured Call Summary, not a full transcript. The AI Receptionist must create/update the summary from the actual conversation where sufficient context exists.

- reason for call;
- relevant business context and key facts;
- needs/problem discussed;
- actions completed by the AI;
- appointments or callback actions;
- open actions and required owner/follow-up;
- urgency/escalation where applicable;
- transfer outcome and relevant handoff state.

The summary must be available to the authorised receiving operator for handoff and remain linked to the appropriate communication timeline after the call, subject to access and retention rules.

24. Call record and lifecycle

Each call must have a unique call/session identifier and an auditable lifecycle. At minimum, store only the operationally necessary/authorised metadata available from the implementation, such as direction, start/end timestamps, duration, status/outcome, routing path, relevant destination, queue/hold events, transfer result, category where known, callback/voicemail actions and technical failure states.

The call record must not default to storing a full transcript. Call records, summaries and communications must follow the platform's access model and privacy/retention controls.

25. Concurrent call architecture - non-negotiable

10 CALLS IS A TEST CASE, NOT A LIMIT

The system must be tested with at least ten genuinely simultaneous calls, but this **MUST NOT** be implemented or interpreted as a maximum of ten callers. The architecture must support scaling to materially higher concurrency by provisioning sufficient telephony, AI and infrastructure capacity. Ten is the minimum launch acceptance/load scenario, not the product capacity ceiling.

Every simultaneous caller must receive a fully isolated call and AI session. Ten or one hundred simultaneous callers may be discussing completely different matters. Supported journeys may occur concurrently, including general questions, Discovery Call booking, AI Scan guidance, support, existing-project matters, human transfer, callback and voicemail.

- No conversational context may cross between calls.
- No verification state may cross between calls.
- No booking/tool action may be attributed to another caller.
- No Call Summary or notification may mix callers.
- No routing state may leak between sessions.
- No caller may receive another caller's audio, data or state.
- The application must not serialise independent calls through a single-conversation bottleneck.
- Under provisioned supported load, speech quality, conversational responsiveness and tool reliability must remain within the accepted production quality standard rather than materially degrading because multiple calls are active.

The requirement is scalable architecture plus correctly provisioned deployed capacity. It is not an unlimited-capacity promise independent of provider/account limits. Developers must document relevant concurrency limits of selected providers/accounts, how capacity is provisioned/monitored, and how IO SKY can scale capacity without rebuilding the application architecture.

26. Mandatory concurrent-load scenarios

Acceptance testing must include at least ten simultaneous live/end-to-end sessions with deliberately different journeys. The test must include a representative mix such as:

- caller asking what IO SKY can do;
- caller discussing an Operations/Cyber/Elite Scan;
- caller booking a Discovery Call;
- existing client/project route;
- support route;
- AI-to-human transfer attempt;
- failed/no-answer transfer recovery;
- callback creation;
- IVR fallback session;
- caller disconnect or another independent supported journey.

The test passes only if session isolation, audio/conversation quality, routing, notifications, summaries, tool actions and backend records remain correct for every caller. Developers must retain test evidence sufficient for IO SKY to verify the result.

The implementation must also be capable of load/capacity testing above ten calls. Production capacity targets must be agreed/provisioned against provider limits; scaling to 50/100+ concurrent sessions must not require redesign of the core session model.

27. Capacity exhaustion and graceful degradation

If a configured/provider capacity limit is reached, the next caller must not be intentionally dropped by application logic without a controlled outcome where the telephony provider permits an alternative. Capacity exhaustion must be logged and alertable.

The system must not conceal capacity saturation by falsely reporting AI/provider availability. Operational monitoring should expose concurrent call count, capacity-related failures and relevant provider errors.

28. Provider/service failure matrix

The development team must implement and test distinct failure paths rather than treating all failures as generic call failure.

Failure	Required principle	Prohibited outcome
Conversational AI unavailable	Use traditional IVR if telephony remains functional.	Caller forced through broken AI or silently dropped.
STT/TTS/AI component failure during call	Controlled recovery/fallback where remaining stack permits.	Pretend the AI heard/performed an action.
Booking/CRM/tool failure	State inability; do not confirm success; offer safe alternative.	False booking/update.
Human destination busy/no-answer	Recover caller; callback-first fallback.	Immediate hang-up or false transfer success.
SIP/routing failure	Try authorised configured fallback where appropriate; log exact failure.	Route to unintended destination.
Notification channel failure	Preserve call/handoff record and other authorised channels.	Corrupt transfer state.
Telephony provider failure	Provider-constrained handling; log/alert when possible.	Fabricated successful call state.

29. No false success / idempotency

Consequential telephony actions must use provider/backend-confirmed state. The system must not say or record that a transfer, callback, voicemail, booking or other action succeeded unless the authoritative backend/provider confirms success.

Retries and webhook/event redelivery must not create duplicate callbacks, duplicate notifications, duplicate booking actions or duplicate call records. Idempotency must be designed into consequential actions.

30. IO SKY business number and SIP identity

The public caller relationship is with the approved IO SKY business number(s). Internal routing to an employee's work/mobile destination must not unnecessarily expose that private/direct number to the caller.

Authorised users with permitted telephony/SIP outbound capability must be able, where technically and legally supported, to place manual business calls using the approved IO SKY caller identity rather than exposing their own personal/direct number.

The architecture must separate business-number configuration from conversation logic so numbers can be migrated or additional approved numbers can be added without rebuilding AI/IVR behaviour.

Future additional/international business numbers may have their own language/routing configuration while using the same controlled telephony foundation.

31. Manual outbound telephony boundary

This specification includes the shared capability required for authorised manual outbound business calling through IO SKY telephony/SIP identity. It does not define automated AI Sales Outbound campaigns, automated prospecting logic or outbound marketing compliance workflows; those belong to a separate system/specification.

Outbound permissions must be explicit. Call direction must be recorded. Applicable recording/transcription state and caller-identity rules must apply equally to authorised outbound calls.

32. Caller identity and verification boundary

Phone number matching may identify a potential Person/Lead/Organisation/Client record and may reduce unnecessary repetition, but it is not sufficient verification for sensitive account or booking actions.

Withheld/unknown caller ID must not block ordinary inbound help. It only limits features that require verified identity or reliable contact data. Sensitive actions remain governed by the verification rules in the AI Receptionist specification.

33. Security, abuse and prompt/social engineering

The telephony layer must tolerate normal frustration and repeated legitimate calls without treating callers as attackers. It must also support reasonable protections against automated abuse, extreme repeated calling, provider-cost abuse and attempts to manipulate protected actions.

DTMF, caller ID, spoken claims or AI instructions cannot grant permissions. Backend authorisation remains authoritative. Telephony inputs must be validated before they trigger consequential actions.

34. Emergency / non-IO-SKY safety matters

IO SKY is not an emergency service. Where a caller describes an immediate personal safety emergency, the system must not present IO SKY support as emergency response. The conversational behaviour is governed by the AI Receptionist Playbook; IVR routes must not misrepresent IO SKY as an emergency service.

35. Super Admin IVR & Telephony Control Center

Super Admin is the launch control authority for IVR and telephony configuration. Normal operational changes must not require code edits.

- business number(s) and applicable configuration;
- AI-first / authorised fallback activation settings;
- Dutch/English IVR prompts and audio;
- menu keys/options/order;
- destinations, routing priorities and fallbacks;
- operator availability and business/exception hours;
- queue settings and hold audio;
- transfer attempt/ring-window configuration;
- callback and voicemail configuration;
- recording/transcription controls and approved notices;
- manual outbound telephony permissions/configuration where applicable;
- notification channel configuration;
- analytics/operational health visibility;
- test/preview and controlled publication of configuration.

36. Configuration lifecycle and safe publication

Material IVR configuration must support a controlled lifecycle comparable to Draft → Test/Preview → Authorised Publish → Active Version, with previous versions retained/auditable and rollback where technically appropriate.

A draft change to routing, recording, menu options or hours must not silently alter the live call path before authorised publication.

Testing must be possible without exposing test routing or test audio to normal production callers.

37. Auditability

Sensitive telephony configuration changes must be auditable. Where applicable the audit record must identify actor, timestamp, resource/configuration changed, previous value/state, new value/state and environment.

At minimum, routing destinations, business hours/availability rules, recording/transcription settings, menu changes, business-number configuration, telephony permissions and publication/rollback actions require audit coverage.

Audit visibility must follow platform permissions. Audit logging itself must not expose secrets unnecessarily.

38. Permissions and Technical Operator

Launch IVR/telephony administration is Super Admin controlled. Future delegation must use explicit granular permissions; a delegated role cannot modify its own privileges.

A Technical Operator is a task-scoped technical role, not a blanket right to telephony/customer data. A Technical Operator may receive only the specific IVR/telephony resources, project/environment and actions necessary for the assigned technical task.

- No automatic access to CRM/client records.
- No automatic access to Call Summaries.
- No automatic access to recordings or persistent transcripts.
- No automatic access to customer communications or commercial context.
- No automatic access to unrelated telephony configuration or secrets.
- If limited sensitive access is strictly necessary for troubleshooting, it must be explicitly authorised, minimised and auditable.

Authorization follows the IO SKY rule: Role + Permission + Scope + Assignment/Ownership + Environment + Resource.

39. Notifications

Telephony notifications must be actionable, permission-aware and linked to the underlying authorised record. A notification never grants access.

- incoming AI-to-human handoff;
- callback required;
- voicemail received;
- failed transfer requiring follow-up;
- potential critical/urgent call where permitted by the AI specification;
- capacity/provider/telephony operational alert for authorised technical/admin users.

Routine successful background events must not create notification spam. Resolved items must not remain presented as unresolved attention.

40. Analytics and operational visibility

Super Admin-facing telephony analytics must include at least the metrics necessary to understand usage, routing and reliability:

- inbound calls and authorised outbound calls where applicable;
- answered/completed/disconnected/failed;
- AI-served calls and AI→IVR fallbacks;
- IVR menu selections;
- queue entries, wait/abandon outcomes where available;
- human transfers attempted/succeeded/failed/no-answer/busy;

- callbacks created/resolved;
- voicemail events;
- call duration;
- technical/provider failures;
- peak/current concurrent call counts where available;
- capacity-related failures or saturation indicators.

No fabricated AI quality score may be presented. Analytics must not unnecessarily duplicate sensitive conversational content.

41. Privacy and data minimisation

Telephony functionality must follow privacy-by-default principles and the applicable IO SKY privacy/DPA/retention framework. The system must collect/store only data needed for the authorised purpose and configured functionality.

External notifications must minimise sensitive content. Recordings/transcripts must not become broadly accessible merely because a user has platform access. Retention/deletion behaviour must be governed by the applicable authorised policy/configuration rather than indefinite storage by default.

42. Accessibility and caller usability

The fallback IVR must use clear, comprehensible prompts, sufficient response time and predictable key behaviour. It must not require the caller to memorise a long list of departments. The launch three-option menu is intentionally compact.

A caller who makes a reasonable mistake must be able to recover. The IVR must not punish slower input with immediate hang-up or repeatedly restart the entire call without need.

43. Acceptance test matrix - mandatory

1. AI available: caller reaches AI Receptionist directly and hears no DTMF menu.
2. AI unavailable before call: caller automatically receives functioning traditional IVR if telephony remains available.
3. AI fails mid-call: controlled recovery occurs where remaining telephony permits.
4. AI recovers while caller is already in IVR: active IVR session remains consistent.
5. Dutch and English fallback paths operate correctly.
6. All three canonical launch menu options route to the configured category/destination.
7. Invalid/no input cannot create an infinite loop.
8. Business hours and actual operator availability are treated separately.
9. Busy/unavailable operator is not pointlessly attempted when effective availability is known.
10. Queue, hold audio and configured queue fallback operate correctly.
11. Hold does not expose receiving-side audio.
12. AI→human transfer creates a compact handoff summary before/as the call is presented.
13. Authorised operator can receive handoff context without already having a laptop dashboard open, through configured channels.
14. Transfer attempt window is configurable and not an unreasonably short hardcoded timeout.
15. Successful transfer is confirmed by authoritative call state.
16. No-answer/busy/technical transfer failure returns caller to a controlled fallback; no silent disconnect.
17. Callback-first fallback creates exactly one callback record and actionable notification.
18. Caller-ID-aware callback confirms existing inbound number rather than needlessly requiring re-entry.
19. Unknown/withheld caller ID still permits ordinary inbound assistance.
20. Voicemail success/failure state is accurate.
21. Recording OFF by default.
22. Recording ON applies configured notice/logging/storage/access behaviour before relevant recording.
23. Persistent transcription is independently controlled from recording and real-time STT.
24. Structured Call Summary is factual, linked correctly and does not default to full transcript.

25. At least ten simultaneous differentiated calls pass with complete session isolation.
26. Ten-call acceptance test is not implemented as a maximum call limit.
27. Load/capacity behaviour above ten can be tested and scaled without redesigning core session architecture.
28. Capacity exhaustion is observable and gracefully handled where provider capability permits.
29. Provider/tool failure does not produce false success.
30. Webhook/retry behaviour does not duplicate callback/notification/action records.
31. IO SKY business caller identity is preserved for authorised manual outbound calls where supported.
32. Internal/direct employee number is not unnecessarily exposed to caller.
33. Super Admin can change normal IVR configuration without code changes.
34. Draft/test changes do not alter live production before authorised publish.
35. Sensitive telephony configuration changes are auditable.
36. Technical Operator cannot see sensitive call/client content without explicit scoped authorisation.
37. Analytics reflect real telephony states and concurrency/capacity events without invented scores.

44. Developer implementation evidence required

Before production acceptance, the development team must provide sufficient evidence for IO SKY to verify the implementation. Evidence must cover:

- architecture/call-flow diagram showing AI-first entry and IVR fallback;
- configured provider/account concurrency limits and scaling method;
- successful ten-simultaneous-call differentiated acceptance test evidence;
- session isolation evidence and representative backend records;
- AI unavailable / mid-call failure / transfer failure demonstrations;
- queue/hold/callback/voicemail demonstrations;
- Super Admin configuration and publication workflow demonstration;
- recording/transcription state demonstration without exposing unnecessary real personal data;
- permission demonstration for Super Admin and scoped Technical Operator;
- analytics/monitoring demonstration including concurrency/capacity visibility;
- known provider constraints and any remaining production prerequisites.

A successful demo of a single happy-path call is not sufficient acceptance evidence.

45. Definition of Done

The IVR & Telephony scope is not Done merely because a telephone number answers. It is Done only when the required AI-first entry, fallback IVR, routing, availability, queue/hold, transfer/handoff, callback, voicemail, business identity, recording/transcription interfaces, session isolation, concurrency, administration, permissions, audit, analytics, failure recovery and mandatory acceptance tests are implemented and demonstrably pass in the intended environment.

Any provider credential, account capacity, DNS/number provisioning, SIP configuration or external dependency that prevents production completion must be explicitly identified as a production prerequisite rather than silently omitted.

46. Non-negotiable summary

- AI Receptionist first; no traditional menu when AI is available.
- Traditional IVR is controlled fallback, not the default caller experience.
- Launch fallback menu has three concise caller-facing options in Dutch and English.
- Super Admin can configure normal IVR/telephony operations without code changes.
- Human business hours and real human availability are separate.
- Queue capability is required and future multi-operator routing is supported.
- AI-to-human handoff provides a compact summary before/as the human receives the call.
- Transfer failure never silently disconnects the caller when a controlled fallback remains possible.
- Callback is preferred before voicemail.

- Caller ID avoids needless re-entry but never substitutes for identity verification.
- Recording is OFF by default; recording/transcription states trigger the correct governed behaviour.
- Call Summary, not full transcript, is the default durable conversational artifact.
- At least ten simultaneous calls must be acceptance-tested; ten is explicitly NOT the maximum.
- The architecture must scale to materially higher concurrency with appropriate provider/infrastructure provisioning.
- Supported concurrent load must preserve session isolation and accepted production quality.
- IO SKY business identity is used for authorised telephony; employee/direct numbers are not unnecessarily exposed.
- Technical Operator access is granular, minimal and auditable.
- No telephony action may be reported successful without authoritative confirmation.
- The developer must demonstrate failure paths, concurrency and administrative control before acceptance.

47. Document control

Document: IO SKY IVR & Telephony Functional Specification

Version: 1.0

Status: Final for Implementation

Owner: IO SKY

Related specifications: AI Receptionist Functional Specification v1.0; AI Receptionist Conversation & Response Playbook v1.0.

Any change that materially alters caller routing, permissions, recording/transcription behaviour, concurrency guarantees, human handoff, business identity or fallback behaviour requires authorised change control.